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Studierichting: Informatica
Oefeningenreeks 2D nr. 10.14

Bereken de volgende limiet:

$$\lim_{x \to 2} \frac{x^3 - x^2 - 8x + 12}{3x^3 - 10x^2 + 4x + 8} = \cancel{\frac{0}{0}}$$

Teller ontbinden:

| | | | | |
|---|---|---|---|---|
| | 1 | $-1$ | $-8$ | 12 |
| 2 | | 2 | 2 | $-12$ |
| | 1 | 1 | $-6$ | 0 |

Noemer ontbinden:

| | | | | |
|---|---|---|---|---|
| | 3 | $-10$ | 4 | 8 |
| 2 | | 6 | $-8$ | $-8$ |
| | 3 | $-4$ | $-4$ | 0 |

$$\Leftrightarrow \lim_{x \to 2} \frac{\cancel{(x-2)}(x^2 + x - 6)}{\cancel{(x-2)}(3x^2 - 4x - 4)} = \cancel{\frac{0}{0}}$$

Teller verder ontbinden:

| | | | |
|---|---|---|---|
| | 1 | 1 | $-6$ |
| 2 | | 2 | 6 |
| | 1 | 3 | 0 |

Noemer verder ontbinden:

| | | | |
|---|---|---|---|
| | 3 | $-4$ | $-4$ |
| 2 | | 6 | 4 |
| | 3 | 2 | 0 |

$$\Leftrightarrow \lim_{x \to 2} \frac{\cancel{(x-2)}(x+3)}{\cancel{(x-2)}(3x+2)} = \frac{5}{8}$$

# References